CSE 211 – One-Page Exam Strategy

From 17 BUET finals (2006-07 → 2023-24, 397 sub-questions). “Recent” = last 5 papers. Section A ≈ regular languages (Hakim Sir); Section B ≈ CFG/PDA/TM/complexity (Atif Sir + Hakim L7–8).

1 · Topics ranked by importance

#	Topic	All-time %	Recent %	Papers /17	Level
1	**DFA design & reading DFAs**	9.9	10.7	17	**MUST**
2	**CFG design**	7.3	8.1	17	**MUST**
3	**Decidability, RE/recursive, halting, diagonalization**	9.3	6.7	17	**MUST**
4	**Complexity: P, NP, NP-complete, Cook, PSPACE**	4.9	8.9	6	**MUST**
5	**Pumping lemma for regular languages**	7.5	6.6	17	**MUST**
6	**Turing machine design / tracing**	8.4	5.6	17	**MUST**
7	**Closure properties of regular languages**	5.1	7.5	11	**HIGH**
8	**NFA / ε-NFA design & concepts**	5.3	7.0	14	**HIGH**
9	**PDA design / tracing**	7.1	5.6	17	**HIGH**
10	**Regular expressions (write / describe / simplify)**	5.3	6.7	15	**HIGH**
11	**TM variants & simulations (multitape, NTM, tracks, computer)**	5.1	5.0	14	**HIGH**
12	**CNF, grammar simplification & CYK**	3.8	5.6	10	**HIGH**
13	**Derivations, parse trees & ambiguity**	5.5	4.1	15	**MEDIUM**
14	**NFA → DFA (subset construction, ε-closure)**	4.0	2.5	13	**MEDIUM**
15	**Pumping lemma for CFLs**	2.8	3.4	8	**MEDIUM**
16	**Basics: languages, ε/∅, role of states, applications**	2.1	3.4	7	**MEDIUM**
17	**RE ↔ FA conversion (RE→NFA, DFA→RE)**	2.5	1.9	8	**MEDIUM**
18	**PDA ↔ CFG, final-state vs empty-stack, DPDA**	3.7	0.6	9	**MEDIUM**

2 · What to read, in this order

No.	Slides	Topic	Practise exactly this (repeated in past papers)
1	Hakim L1–L3	Languages basics → **DFA design**	mod-n / divisible-by-n DFAs, “R resets sum”, substring & run constraints, describe a given DFA
2	Hakim L4	**NFA** + NFA→DFA (subset, ε-closure)	few-state NFAs, conversion from table, why $\emptyset^* = \{\varepsilon\}$
3	Hakim L5	**Regular expressions** + **closure** constructions	write/choose correct RE with reasons; union & cases; flawed star construction; swap-accept fails for NFA
4	Hakim L6	**Pumping lemma (regular)**	prime length, 0^n1^n, squares; “find the flaw in Skye/BLP’s proof”; why $\lvert xy\rvert \le p$
5	Hakim L7	**CFG design** + derivations / **ambiguity**	balanced parens, equal 0s/1s, ≥ three 1s, $a^nb^nc^md^m \cup a^nb^mc^md^n$ + two parse trees
6	Atif L2	**CNF conversion + CYK**	same grammar S→BZ|CY|ZZ|YY every year – just practise the table
7	Hakim L8, Atif L3	**PDA design** (then PDA↔CFG lightly)	$a^ib^jc^k$ with i=j or i=k, $a^nb^mc^md^n$, ww^R, ARITH expressions
8	Hakim L7	**Pumping lemma for CFL**	$a^nb^nc^n$; state & prove the lemma ($\lvert w\rvert \le 2^{n-1}$ bound)
9	Atif L4–L5	**Turing machine design**	bitwise XOR/AND/OR/NOT, equality, 2’s complement on w_1cw_2c using tracks
10	Atif L6	**TM variants**	multi-track one-tape simulates multitape; NTM→DTM; polynomial-simulation true/false
11	Atif L7	**Undecidability**	L_d not RE; halting RE-but-not-recursive; universal language; one example per language class
12	Atif L8–L9	**Complexity**	Cook’s theorem, NPC ⇒ P=NP, Venn of P/NP/PSPACE, PSPACE = NPSPACE (Savitch)

Short on time? Do steps 5–12 first: Section B has used nearly the same 9 questions since 2019 (CFG · ambiguity · CYK · PDA · CFL pumping · TM design · TM variants · undecidability · complexity). Then steps 1–4 for Section A’s compulsory Q1. Finish by solving the 2023-24, 2022-23 and 2021-22 papers under time. **Skim only:** PDA↔CFG proofs, DFA→RE state elimination, Rice’s theorem.